VERIFICATION CORE REPORT

Topic: **Java 21 Virtual Threads & Structured Concurrency** | College: **VIT**
Tech: **Java 21.0.6** | Professor: **sam** | Deck ID: **67e0fec7** | Date: 2026-10-17 12:55:47
Docker Image Digest: **eclipse-temurin:21-jdk-alpine@sha256:7b5d1e4392a874f6e21b03e2c34a6e8b7c439f018e652a129efb923415cf98a1**

Metric	Value	Metric	Value
Found vs Executed	19 found / 19 run	Unverified (Not Run)	0
Snippets Verified	0/19	Auto-Healed	0
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	9333%
Template QA Passed	5/11	Docker Backend Runs	6

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	634.675698852539ms	`Result: Verified Output`
Slide 4	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	491.2958676300049ms	`Result: Verified Output`
Slide 5	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	495.8251449127197ms	`Result: Verified Output`
Slide 6	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	491.1254718322754ms	`Result: Verified Output`
Slide 7	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	496.1372100830078ms	`Result: Verified Output`
Slide 8	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	506.5954262695312ms	`Result: Solution Verified`
Slide 9	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	470.1246460723877ms	`Result: Verified Output`
Slide 3	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		`(none)`

Slide 4	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		(none)
Slide 5	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		(none)
Slide 6	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		(none)
Slide 7	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		(none)
Slide 9	java 21.0.6 (docker)	VERIFIED LOW ASSURANCE		(none)
Slide 3	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5641642169952392	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 4	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5347622799606323	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 5	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5288382854507446	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 6	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5585615186691284	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 7	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5508246784210205	Java 21 Virtual Threads & Structured Concurrency: Concept Executed Successfully
Slide 9	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	5521CE6584167480	Quick Knowledge Check Executed Successfully

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-8-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	4/5	80.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-8-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 4	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 5	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 6	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 7	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.
Slide 11	COMPLIANT	All bounding boxes and fonts compliant.